Byung-cheol Kang

Genomics · Bacteriophages · Research Pipelines

A bioinformatics researcher and developer connecting biological questions with analysis design and interpretation. My work extends from microbes to humans, integrating diverse analytical approaches and building research pipelines and databases. AI-powered coding accelerates my research and amplifies my ability to automate workflows and develop analytical tools.

CURRENT AND RECENT ROLES

MicrobiotiX · Bioinformatician — 2026.07 - PRESENT

- Develop bacterial genome analysis workflows covering quality control, annotation, typing, antimicrobial resistance, and prophage detection to support production-strain assessment.
- Integrate phage genome quality, termini and packaging, taxonomy, lifestyle predictions, and tail-protein evidence into reports with traceable conclusions.
- Develop and extend analysis libraries and command-line tools combining experimental host-range matrices with sequence and protein features for clustering comparisons and cocktail candidate selection.
- Build and maintain a Klebsiella pneumoniae tail/RBP reference database linking sequences to K-type and KL-type evidence, with source citations and separate experimental and predicted labels.
- Develop in silico design tools that evaluate tail-protein candidates for specified capsule targets and generate candidate design sequences and review reports.
- Compare protein sequences, domains, predicted structures, and reference structures to support RBP annotation and assess the structural plausibility of replacement and fusion candidates.

Wittgenbio · Computational Biology Research — 2026.07.01 - 2026.10.01

- Built a colorectal cancer (CRC) single-cell atlas and performed cell-type annotation.
- Annotated externally supplied normal-tissue atlas data and compared tissue-specific expression to evaluate candidate therapeutic targets.
- Contributed to virtual-cell research through computational cell-state representation, model evaluation, and biologically grounded interpretation.

Past platform evidence. The integrated workflow and operational results in this portfolio describe work performed at Sanigen (2020.03 - 2025.10). They are not presented as MicrobiotiX or Wittgenbio assets or current-company outcomes.

PROJECT 01 · PLATFORM

Integrated Microbial Genomics Platform

PROBLEM

Separate assay scripts, environments, quality checks, server paths, and reporting conventions made analysis difficult to reproduce and review consistently.

SYSTEM DESIGN

Entry point	One Snakemake dispatcher selects 16S, ITS, WGS, or Hybrid profiles.
Execution	Server-specific YAML configuration with Conda, Docker, or Apptainer and pinned BioContainers.
Quality	Input validation, raw-read QC, assembly/analysis checks, review gates, and retained logs.
Delivery	Custom Python/R processing, visualizations, HTML reports, and packaged outputs.

ACTIVE TOOL CHAINS

Profile	Core processing	QC / interpretation
16S / ITS	QIIME 2, DADA2, SILVA classifier, MAFFT, FastTree	FastQC, MultiQC, diversity metrics, custom R/Python reports
WGS	Trimmomatic, SPAdes, Bowtie2, samtools, Prokka	QUAST, BUSCO, Kraken2, Abricate/ CARD/VFDB, eggNOG-mapper
Hybrid	NanoFilt, NanoPlot, Flye or Unicycler, BWA, Pilon, minimap2	QUAST, BUSCO, Prokka, CGView, BLASTn, Abricate, eggNOG-mapper

EVIDENCE BOUNDARY

Only tools connected to active Snakemake includes and rules are listed as active. Configured or documented but inactive components are not represented as production use.

PROJECT 02 · OPERATIONS

Analysis Operations and Quality Assurance

OPERATING MODEL

Question
biological scope

→

Execution
workflow + QC

→

Delivery
interpretation + provenance

RESPONSIBILITIES AT SANIGEN

- Designed and maintained a shared analysis platform for 16S, ITS, WGS, and hybrid assembly, standardizing execution, quality control, and reporting.
- Performed microbial genomics and metagenomics analyses, led technical review, and coordinated team assignments, project schedules, and client communication.
- Contributed comparative-genomics and NGS analyses to foodborne-pathogen panel development, supporting two peer-reviewed publications and a joint patent.

VERIFIED LOWER BOUND OF ARCHIVED WORK

28
WGS order codes

21
Hybrid order codes

19
16S order codes

18
Shotgun order codes

7
RNA order codes

3
ITS order codes

2
Other order codes

These 98 distinct order codes are a conservative lower bound derived from archived communications. They are not sample counts, revenue, or an independently audited total workload.

From Genome Differences to Assay Candidates

COMPARATIVE-GENOMICS MARKER DISCOVERY

Objective. Identify a region that distinguishes a target group while remaining practical for downstream marker or assay development.

Approach. Curate complete genomes, compare whole-genome and pan-genome context, narrow candidate intervals, and review conservation, specificity, and downstream constraints.

Outcome. Prioritized an approximately **17 kb candidate region** for follow-up, with traceable reasoning from genome selection to practical assay constraints.

FOODBORNE-PATHOGEN NGS PANEL

- Contributed bioinformatics and comparative-genomics analysis to marker-oriented panel development and evaluation.
- Supported interpretation of NGS outputs across food and environmental sample contexts.
- Collaborative outcomes included two peer-reviewed papers in 2023 and one joint patent contribution in 2022.

BOUNDARIES

> Computational prioritization does not establish diagnostic sensitivity or specificity. Publication and patent records document collaborative outcomes, not sole authorship or universal performance across unevaluated matrices.

SELECTED EVIDENCE

- **Frontiers in Microbiology** 14 (2023): agricultural wastewater pathogen panel.
- **Journal of Microbiology and Biotechnology** 33(1) (2023): fermented-food pathogen panel.
- Joint patent contribution (2022): NGS-based primer set and foodborne-bacteria detection method.

PROJECT 05 · CURRENT ROLE CASE STUDY

Klebsiella Phage Genomics and Tail-RBP Design

GENOME EVIDENCE WORKFLOW

Host + phage genomes
QC + annotation

→

Independent evidence
taxonomy + lifestyle + termini

→

Tail interpretation
KL target + structure

CONTRIBUTION AND EVIDENCE

- Integrated CheckV, Pharokka, Phold, BACPHLIP, PhaTYP, PhaGCN, and PhageTerm outputs while preserving the evidence behind each conclusion.
- Built a tail/RBP and KL-type reference resource with source traceability, confidence labels, and experimentally supported targets explicitly flagged.
- Cross-validated candidate depolymerases with DposFinder, SpikeHunter, DepoScope, sequence search, domain boundaries, and protein-structure evidence.
- Used ProstT5 and Foldseek with experimental PDB references to identify structurally conserved proteins at sequence identities below ordinary search sensitivity.

ANONYMIZED CASE-STUDY RESULTS

- Four genomes converged on lytic **Przondovirus** classifications and short DTR packaging with 180-183 bp terminal repeats.
- Three capsule targets matched experimentally supported references at 95.4-98.8% sequence identity; unsupported cases remained unresolved rather than receiving forced labels.
- One design candidate passed genus, attachment, target-label, experimental-structure, and predicted-structure criteria; its 129-aa anchor alignment spanned 99.2% of the region.

DESIGN OUTPUT AND BOUNDARY

The design bench supports domain swaps, full-module replacements, and tandem additions, with genome FASTA, module multi-FASTA, GenBank, and protein FASTA export. Host K-locus assignments, predicted RBP folding, adsorption, host range, and lytic activity still require bacterial-genome and wet-lab validation.

PROJECT 06 · WITTGENBIO RESEARCH

CRC Atlas and Cell-Type Annotation

CRC SINGLE-CELL ATLAS WORKFLOW

Single-cell datasets
QC + metadata

→

Integrated CRC atlas
harmonization + clustering

→

Cell-type annotation
labels + validation

ANALYTICAL RESPONSIBILITIES

- Built a colorectal cancer single-cell atlas by integrating datasets across studies and performing cell-type annotation.
- Maintained traceable metadata and reviewed annotation consistency and tumor/normal label discrepancies.

NORMAL-TISSUE ATLAS ANNOTATION AND EXPRESSION

- Annotated externally supplied normal-tissue atlas data and compared tissue-specific expression.
- Evaluated candidate therapeutic targets against normal-tissue expression while accounting for sequencing-depth differences and annotation coverage.

VIRTUAL-CELL WORKFLOW

Cellular data
question + scope

→

Cell model
representation + fitting

→

Evaluation
state + response + biology

EVALUATION PRINCIPLES

- Separate model fitting from evaluation data across relevant cell states, conditions, and perturbations.
- Evaluate predicted states or responses against measurable biological evidence, not latent-space appearance alone.
- Report underrepresented tissues, cell populations, perturbations, and experimental conditions.

PUBLIC EVIDENCE BOUNDARY

Cell-type annotations and normal-tissue expression comparisons depend on data quality, annotation coverage, and biological context. Virtual-cell predictions remain research hypotheses until externally or experimentally validated. Private data, implementation details, and performance metrics are not disclosed here.

PROJECT 07 · INDEPENDENT R&D PROTOTYPE

Phage Lifestyle Evidence Prototype

RESEARCH QUESTION

Can existing tools be combined into an end-to-end system that distinguishes lytic and lysogenic phages while preserving conflicting evidence and uncertainty?

EVIDENCE DESIGN

Quality context	CheckV completeness and contamination context
Annotation	Pharokka and Phold; integrase, repressor, excision, and lysis evidence
Classifiers	BACPHLIP and PhaTYP as independent prediction evidence
Decision layer	Agreement/confidence table with unresolved cases routed to manual review

HOST-RANGE EXTENSION

A related hypothesis-ranking layer considers LPS and outer-membrane gene variation together with phage receptor and annotation evidence. It prioritizes candidates for experiment; it does not prove adsorption, infection, or lytic activity.

PUBLICATION-GRADE NEXT STEPS

- External, leakage-controlled benchmark with known lifestyle labels.
- Comparison against individual tools and transparent simple baselines.
- Sensitivity, specificity, MCC, calibration, and quality-stratified performance.
- Ablation analysis for gene markers, classifiers, and quality context.
- Experimental validation for host-range candidate claims.

CONTACT

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